

Fold It Yourself: a lattice protein

Watching a sequence of fourteen letters decide a shape, and finding out exactly how far that claim goes.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/fold-it-yourself/

1 Before you open anything

Answer from what you already know. The point is to have a number on paper that the demonstration will either confirm or contradict.

PREDICTION

In this model a hydrophobic (H) residue that touches another H residue — without the two being neighbours in the chain — counts as one **H-H contact**, worth -1ϵ of energy. The folded molecule has six such contacts.

Now heat it. At the hottest setting the page allows, $T^* = 3.00$, the chain is thoroughly denatured. What number will the readout **Average H-H contacts** show? Commit to a value.

Write one sentence saying why you chose that number.

2 Read the folded molecule

Open the demonstration. It starts on the **Designed** sequence, already sitting in its lowest-energy shape.

1. In the **Primary structure** panel, press **Designed**. Copy the fourteen letters shown below the buttons into the boxes in question **a**.
2. In the **Fold it** panel, press the button whose label begins **Show** and ends **lowest-energy shape**. The molecule on screen is now the best fold this sequence has.
3. Check that the **Temperature T^*** slider reads **0.20**. Set it there if it does not.
4. Fill in the table from the readouts. Each row names the label exactly as it appears on screen.

Readout	Value
H-H contacts now	
Energy now	
Lowest energy possible	
Shapes tied at that energy	
Sides facing water, H average	
Sides facing water, P average	
Greasy residues left out in the water	
Melting temperature	

a. Write out the sequence, and count how many residues are H and how many are P.

b. The two “sides facing water” numbers are averages out of a possible 4. Say in one sentence what the difference between them tells you about where the H residues have ended up, and where the P residues have ended up.

c. Nothing in this model instructs a residue to go inside or outside. The only rule is that an H-H contact lowers the energy. Explain how that single rule produces the arrangement you described in **b**.

d. Shapes tied at that energy counts rotations and mirror images once. State what the value you recorded means for the question “does this sequence have one structure?”

The demonstration is not fitting a curve or guessing. It scores all 220,375 ways a 14-bead chain can sit on

a square grid and reports the best of them. Within the rules of the model, these numbers are exact.

3 Melting it

One control, varied systematically. Keep the **Designed** sequence loaded throughout. The two readouts you are recording are equilibrium values — they update the moment you move the slider, whether or not the animation is running.

1. Drag the **Temperature T^*** slider to each value in the table. It moves in steps of 0.01 between 0.05 and 3.00, so every value listed is reachable.
2. At each stop record **Molecules in the lowest-energy shape** and **Average H–H contacts**.

T^*	Molecules in the lowest-energy shape (%)	Average H–H contacts
0.05		
0.20		
0.30		
0.40		
0.50		
1.00		
2.00		
3.00		

a. Between which two of your temperatures does the folded fraction fall through 50%? Compare that interval with the **Melting temperature** readout.

b. Describe, in your own words, the shape of the folded-fraction curve: what it does over most of the temperature range, and what it does near the melting temperature.

c. Go back to your prediction in part 1. What did the readout actually say at $T^* = 3.00$? If your number was different, name the assumption behind it that turned out not to hold.

d. Under **Average H–H contacts** the page prints a *random-coil floor*. Record that number, and explain why a chain thrown onto the grid at random still makes some contacts even though nothing is holding it together.

e. Leave the slider at 3.00 and press **Run at this T**. Watch the picture. Does the backbone ever break? Then slide back down to 0.20 with the simulation still running and describe what happens.

This melting curve is an equilibrium, so cooling really does refold the chain. In a crowded cell or a hot beaker, unfolded chains usually stick to one another and aggregate first, and aggregation is not reversible — which is why a boiled egg stays boiled. That is a traffic problem, not a rule of physics.

4 One substitution at a time

A tempting shortcut: judge a mutation by whether it changes the energy. Three presets are enough to show what that shortcut misses. Each preset snaps the molecule to a lowest-energy shape, so the picture you see is the fold, not a half-finished attempt.

1. Press **Designed** and fill in the first row.
2. Press **One letter → new shape** (residue 6, H → P) and fill in the second row. Note whether the picture is the same fold or a different one.
3. Press **Sickle-style patch** (residue 2, P → H) and fill in the third row.

Preset	Lowest energy possible	Shapes tied	Sides facing water, H average	Greasy residues in the water	Melting temperature
Designed					
One letter → new shape					
Sickle-style patch					

a. Compare rows 1 and 3. The energy and the fold are the same. State precisely what the demonstration shows *did* change, and explain why that change could matter to a cell even though the molecule has the same shape.

b. The melting temperature in row 3 is not the same as in row 1, even though the lowest-energy shape is identical. Suggest a reason a substitution can leave the best fold untouched and still lower the temperature at which half the molecules stay folded.

c. *Working backwards.* Press **Designed** again. Clicking any numbered residue in the **Primary structure** panel switches it between H and P, and the readouts recompute at once. Find **one** click that drops **Lowest energy possible** from -6 to -4. Record the residue number, the change you made, and what **Shapes tied at that energy** becomes.

d. Press **Designed** to reset. Now find **one** click that leaves **Lowest energy possible** at -6 but makes **Shapes tied at that energy** greater than 1. Record it.

e. The sequence you found in **d** is just as stable as the original by the energy readout. Explain why it is nevertheless a worse protein, using the number of tied shapes in your answer.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, check yourself, and write down where any disagreement came from.

1. Colour the square grid like a chessboard. Residue 1 sits on a dark square; every step along the chain moves to a square of the other colour.

(i) Explain why two residues can be next to each other on the grid only if the number of chain steps between them is *odd*.

(ii) The designed sequence is **HPHPPHPHPPHPPH**, with H at positions 1, 3, 6, 8, 11 and 14. Using the rule from (i), and remembering that residues joined in the chain do not count as a contact, list every pair of H residues that could *possibly* form an H–H contact. How many pairs are there?

(iii) The demonstration reports a lowest energy of -6ϵ , not the number you got in (ii). Explain why your count is an upper bound rather than the answer.

2. Each of the 14 beads has 4 sides. Two beads sitting next to each other on the grid hide one side from the water each. In the folded designed molecule the beads touch in exactly 19 places: 13 peptide bonds along the backbone and 6 H–H contacts.

(i) Calculate how many bead-sides in total are facing water.

(ii) There are 6 H beads and 8 P beads. Use your part-2 readings for **Sides facing water, H average** and **Sides facing water, P average** to get the same total a second way, and check that the two answers agree.

3. Two sequences differ at a single residue. Both have a lowest energy of -6ϵ , both have exactly one shape tied at that energy, and the two shapes are the same shape. Explain, without doing any arithmetic, how such a substitution can still cause a disease. Name the property that changed and the physical consequence it has when many copies of the molecule are present at once.

4. This 14-bead chain has 2.2×10^5 possible shapes. A 100-residue chain on the same flat grid has roughly 10^{42} .

(i) If a chain could test one shape every 10^{-13} s, how many seconds would it take to test all 10^{42} ? Compare that with the age of the universe, about 4×10^{17} s.

(ii) Real proteins of that size fold in milliseconds. Explain in one or two sentences what your answer to (i) rules out about how folding works, and what the demonstration shows happening instead.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet uses.

- Press **Stretch it out**, then drag the beads on the picture with your finger or mouse and try to reach -6ϵ by hand. The chain will not let you break it or overlap it. How close can you get, and what makes the last contact hard to find?
- Press **Anneal (fold it)** repeatedly. Each press uses a new random seed and takes a different path. Keep a tally of how often it finishes at -6ϵ and how often it stops one contact short. A run that stops short is a misfold — a chain trapped in a dead end because it cooled faster than it could rearrange. Real cells employ chaperones for exactly this.
- Press **Random sequence**, a sequence nobody selected. Read **Shapes tied at that energy**, then press the lowest-energy-shape button several times to step through them. Anneal a few times and see which one you land on. What does this sequence do to the claim that primary

structure determines tertiary structure — and what does that suggest about how real proteins came to be exceptions?

- Build a sequence with only one H residue, or with none, by clicking residues. Read what the melting curve says, and work out why there is nothing left to melt.